

06.03 Synthetic Division

Concept summary and key formulas

Key ideas

This subsection focuses on polynomial structure and equations.

Look first for structure: degree, factor pattern, roots, or graph behavior.

Key formulas

$$a^2 - b^2 = (a - b)(a + b)$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$f(a) = 0 \Rightarrow (x - a) \text{ is a factor}$$